FYS-STK August 17, 2026

1. input vector

$$x = [x_0, x_1, x_2 --, x_{n-1}]^T$$

$$y = [y_0, y_1, y_2, --, y_{n-1}]^T$$

output

$$x \in \mathbb{R}^n$$

$$y \in \mathbb{R}^n$$

2 Making a model

$$y(x) \simeq f(x;\Theta)$$

parameters

$$\Theta = [\Theta_0, \Theta_1, \Theta_2, \ldots \Theta_{p-1}]$$

$$f(x_i;\Theta) = \sum_{j=0}^{m-1} x_i^j \Theta_j$$

linea $f(x_i;\theta) = \Theta_0 + \Theta_1 x_i^1$

polynomial of degree $n-1$

$$y_0 = c_0 + c_1 x_0^1 + c_2 x_0^2 + \cdots + x_0^{n-1} c_{n-1}$$

$$y_1 = c_0 + c_1 x_1^1 + c_2 x_1^2 + \cdots x_1^{n-1} c_{n-1}$$

$\vdots$

$$y_{n-1} = c_0 + c_1 x_{n-1}^1 + \cdots + x_{n-1}^{n-1} c_{n-1}$$

$$\vec{y} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ \vdots \\ \vdots \\ y_{m-1} \end{bmatrix} = X \cdot \begin{bmatrix} \epsilon_0 \\ \epsilon_1 \\ \vdots \\ \vdots \\ \epsilon_{m-1} \end{bmatrix}$$

matrix

$$= \boxed{X \cdot \vec{\epsilon}}$$

linear dependence

$$X = \begin{bmatrix} 1 & x_0^1 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1^1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & \vdots & & & \\ \vdots & \vdots & & & \\ 1 & x_{n-1}^1 & & \cdots & x_{n-1}^{n-1} \end{bmatrix}$$

Design / feature matrix

Define polynomial of degree $p-1$

$$f(x_i;\theta) = \sum_{j=0}^{p-1} x_i^j \, \theta_j$$

$$\vec{\theta} \in \mathbb{R}^p$$

$$\vec{x} \in \mathbb{R}^n, \quad \vec{y} \in \mathbb{R}^n$$

$$X \in \mathbb{R}^{n \times p}$$

$$p = 3$$

$$f(x_i ; \theta) = \theta_0 + \theta_1 x_i^1 + \theta_2 x_i^2$$

$$X = \begin{bmatrix} 1 & x_0^1 & x_0^2 \\ 1 & \vdots & \\ \vdots & \vdots & \\ \vdots & \vdots & \\ 1 & x_{n-1}^1 & x_{n-1}^2 \end{bmatrix}_{n \times 3}$$

$$X \in \mathbb{R}^{n \times 3}$$

specific feature

③ Assessing the model

$\Rightarrow$ Loss function / cost / Risk / ...

$$\tilde{y}_i = f(x_i ; \theta)$$

$$\vec{\tilde{y}} = [\tilde{y}_0, \tilde{y}_1, \tilde{y}_2, \cdots \tilde{y}_{n-1}]$$

$$\vec{\tilde{y}} \in \mathbb{R}^n$$

suggestions for functions which assess if $\tilde{y}$ is a good or bad model

$$C(y;\theta) = \sum_{i=0}^{n-1} |y_i - \tilde{y}_i|$$

$$\vec{\nabla}_\theta C(y;\theta) = 0$$

MSE

$$C(y;\theta) = \frac{1}{n}\sum_{i=0}^{n-1}\left(y_i - \tilde{y}_i\right)^2$$

$$= \frac{1}{n}\sum_{i=0}^{n-1}\left(y_i - \sum_{j=0}^{p-1} x_i^j \theta_j\right)^2$$

$$= \frac{1}{n}\sum_{i=0}^{n-1}\left(y_i - \theta_0 - \theta_1 x_i^1 - \dots - \theta_{p-1} \times x_i^{p-1}\right)^2$$

$$\frac{\partial C(y;\theta)}{\partial \theta_j} = 0$$

$$C(y;\theta) = \frac{1}{n} \| \vec{\underline{y}} - \underset{(n \times p) \cdot \times p}{\underline{X}\vec{\theta}} \|_2^2$$

$$\|x\|_2 = \sqrt{\sum_{i=0}^{n-1} x_i^2}$$

$$C(y;\theta) = \frac{1}{n} (\vec{y} - X\vec{\theta})^T (y - X\vec{\theta})$$

$$\hat{\epsilon} = \left(X^T X\right)^{-1} X^T y$$